

Tugas 1

Nama : Adi Surya Fangidae

NIM : 055378839

Prodi : Sistem Informasi

1.

Diketahui matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$;

$$B = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Tentukan $I + BA - B^T A^T$!

a) hitung BA :

$$B = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} ; A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow (3)(4) + (-2)(-2) + (0)(-2) = 12 + 4 + 0 = 16$$

$$\Rightarrow (3)(0) + (-2)(1) + (0)(0) = 0 - 2 + 0 = -2$$

$$\Rightarrow (3)(1) + (-2)(0) + (0)(1) = 3 + 0 + 0 = 3$$

$$\Rightarrow [16, -2, 3]$$

$$\begin{aligned} \text{c)} \quad (-2)(4) + (3)(-2) + (0)(-2) &= -8 - 6 + 0 \\ &= -14 \end{aligned}$$

$$\begin{aligned} \text{d)} \quad (-2)(0) + (3)(1) + (0)(0) &= 0 + 3 + 0 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad (-2)(1) + (3)(0) + (0)(1) &= -2 + 0 + 0 \\ &= -2 \end{aligned}$$

$$\therefore [-14, 3, -2]$$

$$\begin{aligned} \text{f)} \quad (0)(4) + (0)(-2) + (5)(-2) &= 0 + 0 - 10 \\ &= -10 \end{aligned}$$

$$\begin{aligned} \text{g)} \quad (0)(0) + (0)(1) + (5)(0) &= 0 + 0 + 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{h)} \quad (0)(1) + (0)(0) + (5)(1) &= 0 + 0 + 5 \\ &= 5 \end{aligned}$$

$$\therefore [-10, 0, 5]$$

Maka hasil dari BA adalah :

$$BA = \begin{bmatrix} 16 & 2 & 3 \\ -14 & 3 & -2 \\ -10 & 0 & 5 \end{bmatrix}$$

b) Hitung $B^T A^T$

Karena B sudah simetris, $B = B^T$. Setarangkan ubah matrix A ke bentuk A^T :

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 4 & -2 & -2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Mencari BA^T

$$B = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}; A^T = \begin{bmatrix} 4 & -2 & -2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \stackrel{1}{\uparrow} \\ (=) (3)(4) + (-2)(0) + (0)(1) &= 12 + 0 + 0 \\ &= 12 \end{aligned}$$

$$\begin{aligned} \stackrel{2}{\uparrow} \\ (=) (3)(-2) + (-2)(1) + (0)(0) &= -6 - 2 + 0 \\ &= -8 \end{aligned}$$

$$\begin{aligned} \stackrel{3}{\uparrow} \\ (=) (3)(-2) + (-2)(0) + (0)(1) &= -6 + 0 + 0 \\ &= -6 \end{aligned}$$

$$=) [12, -8, -6]$$

$$\begin{aligned} (=) (-2)(4) + (3)(0) + (0)(1) &= -8 + 0 + 0 \\ &= -8 \end{aligned}$$

$$\begin{aligned} (=) (-2)(-2) + (3)(1) + (0)(0) &= 4 + 3 + 0 \\ &= 7 \end{aligned}$$

$$\begin{aligned} (=) (-2)(-2) + (3)(0) + (0)(1) &= 4 + 0 + 0 \\ &= 4 \end{aligned}$$

$$=) [-8, 7, 4]$$

$$\begin{aligned} \text{a)} \quad (0)(4) + (0)(0) + (5)(1) &= 0 + 0 + 5 \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad (0)(-2) + (0)(1) + (5)(0) &= 0 + 0 + 0 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{c)} \quad (0)(-2) + (0)(0) + (5)(1) &= 0 + 0 + 5 \\ &= 5 \end{aligned}$$

$$\text{d)} \quad [5, 0, 5]$$

$$\text{Maka } B^T A^T = \begin{bmatrix} 12 & -8 & -6 \\ -8 & 7 & 4 \\ 5 & 0 & 5 \end{bmatrix}$$

c) Mencari Hasil dari $BA - B^T A^T$:

$$BA = \begin{bmatrix} 16 & -2 & 3 \\ -14 & 3 & -2 \\ -10 & 0 & 5 \end{bmatrix} ; B^T A^T = \begin{bmatrix} 12 & -8 & -6 \\ -8 & 7 & 4 \\ 5 & 0 & 5 \end{bmatrix}$$

$$\text{a)} \quad 16 - 12, \quad (-2) - (-8), \quad 3 - (-6) \Rightarrow 4, 6, 9$$

$$\text{b)} \quad (-14) - (-8), \quad 3 - 7, \quad (-2) - 4 \Rightarrow -6, -4, -6$$

$$\text{c)} \quad -10 - 5, \quad 0 - 0, \quad 5 - 5 \Rightarrow -15, 0, 0$$

$$\text{Maka } BA - B^T A^T = \begin{bmatrix} 4 & 6 & 9 \\ -6 & -4 & -6 \\ -15 & 0 & 0 \end{bmatrix}$$

d) Hitung $I + BA - B^T A^T$:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ; BA - B^T A^T = \begin{bmatrix} 4 & 6 & 9 \\ -6 & -4 & -6 \\ -15 & 0 & 0 \end{bmatrix}$$

e) $1 + 4$, $1 + (-4)$, $1 + 0$ $\Rightarrow 5$, -3 , 1

Maka hasil $I + BA - B^T A^T = \begin{bmatrix} 5 & 6 & 9 \\ -6 & -3 & -6 \\ -15 & 0 & 1 \end{bmatrix}$

2. Cari matrix x pada matrix berikut :

$$\begin{bmatrix} 5 & 4 \\ 1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} x = \begin{bmatrix} 3 & -1 \\ -3 & 2 \end{bmatrix}$$

e) Misalkan tiap matrix diberi nama :

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} \quad C = \begin{bmatrix} 3 & -1 \\ -3 & 2 \end{bmatrix}$$

Maka memperoleh persamaan : $A - Bx = C$;
Ubah ruas pers. tersebut $= A - C = Bx$

e) hitung $A - C$:

$$A - C = \begin{bmatrix} 5 & 4 \\ 1 & 8 \end{bmatrix} - \begin{bmatrix} 3 & -1 \\ -3 & 2 \end{bmatrix}$$

$$5 - 3, \quad 4 - (-1) \Rightarrow 2, 5$$

$$1 - (-3), \quad 8 - 2 \Rightarrow 4, 6$$

$$A - C = \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix}$$

*) Dari persamaan sebelumnya $Bx = A - C$, maka diperoleh :

$$\begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} x = \begin{bmatrix} 2 & 5 \\ 4 & 6 \end{bmatrix} \Rightarrow B^{-1}$$

Karena matrix kedua ruas tersebut sama persis, yang merupakan sifat $IA = A$. Maka x merupakan Matrix Identitas bentuk 2×2 .

3. ubah matrix $\begin{bmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 1 & 4 \\ 3 & 7 & 5 & 8 \end{bmatrix}$ menjadi matrix eselon :

$$\Rightarrow R_1 \leftrightarrow R_1 \leftrightarrow R_2 = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 2 & 4 & -2 & 6 \\ 3 & 7 & 5 & 8 \end{bmatrix} \quad (1)$$

$$\begin{aligned} \Rightarrow R_2 &\leftarrow R_2 - 2R_1; \\ R_3 &\leftarrow R_3 - 3R_1 \end{aligned} = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & -2 & -4 & -2 \\ 0 & -2 & 2 & -4 \end{bmatrix} \quad (2)$$

$$b) R_2 \leftarrow -\frac{1}{2} R_2 = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & -2 & 2 & -4 \end{bmatrix} \quad (3)$$

$$c) R_3 \leftarrow R_3 + 2R_2 = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 6 & -2 \end{bmatrix} \quad (4)$$

$$d) R_3 \leftarrow \frac{1}{6} R_3 = \begin{bmatrix} 1 & 3 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -\frac{1}{3} \end{bmatrix} \quad (5)$$

Terpenuhi sebagai
Matrix eselon.

$$4. \quad A = \begin{bmatrix} 5 & 9 \\ 7 & 3 \end{bmatrix} \quad ; \quad kI = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$e) d. (A - kI) : \dots$$

$$A - kI = \begin{bmatrix} 5 & 9 \\ 7 & 3 \end{bmatrix} - \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \quad (:) \quad \begin{matrix} 5-k, & 9-0 \\ 7-0, & 3-k \end{matrix}$$

$$A - kI = \begin{bmatrix} 5-k & 9 \\ 7 & 3-k \end{bmatrix}$$

$$\Rightarrow \det(A - KI) = (5-k)(3-k) - 9 \cdot 7$$

$$\downarrow \qquad \qquad \qquad \downarrow$$

$$0 = (15 - 5k - 3k + k^2) - 63$$

$$0 = k^2 - 8k + 15 - 63$$

$$0 = k^2 - 8k - 48$$

$$(k+4)(k-12) = 0 \quad \rightarrow \text{Faktorkan } (k = -4; k = 12)$$

Maka nilai k yang memenuhi : 12 dan -4

5.

$$\text{Matrix } P = \begin{bmatrix} 3 & 1 & -2 \\ 5 & -2 & -3 \\ 2 & 2 & 3 \end{bmatrix}$$

eliminasi Gauss-Jordan :

$$\begin{aligned} \Leftrightarrow R_2 &\leftarrow R_2 - R_1 ; \\ R_3 &\leftarrow R_3 - R_1 \end{aligned} \quad = \begin{bmatrix} 3 & 1 & -2 \\ 2 & -3 & 1 \\ -1 & 1 & 5 \end{bmatrix} \quad (1)$$

$$\begin{aligned} \Leftrightarrow R_1 &\leftrightarrow R_3 ; \\ R_1 &\leftarrow -R_1 \end{aligned} \quad = \begin{bmatrix} 1 & -1 & -5 \\ 2 & -3 & 1 \\ 3 & 1 & -2 \end{bmatrix} \quad (2)$$

$$\begin{aligned} \Leftrightarrow R_2 &\leftarrow R_2 - 2R_1 ; \\ R_3 &\leftarrow R_3 - 3R_1 \end{aligned} \quad = \begin{bmatrix} 1 & -1 & -5 \\ 0 & -1 & 11 \\ 0 & 4 & 13 \end{bmatrix} \quad (3)$$

$$\Rightarrow R_2 \leftarrow -R_2 = \begin{bmatrix} 1 & -1 & -5 \\ 0 & 1 & -9 \\ 0 & 4 & 13 \end{bmatrix} \quad (4)$$

$$\Rightarrow R_3 \leftarrow R_3 - 4R_2 = \begin{bmatrix} 1 & -1 & -5 \\ 0 & 1 & -9 \\ 0 & 0 & 49 \end{bmatrix} \quad (5)$$

$$\Rightarrow R_3 \leftarrow \frac{1}{49} R_3 = \begin{bmatrix} 1 & -1 & -5 \\ 0 & 1 & -9 \\ 0 & 0 & 1 \end{bmatrix} \quad (6)$$

$$\begin{aligned} \Rightarrow R_2 &\leftarrow R_2 + 9R_3 ; \\ R_1 &\leftarrow R_1 + 5R_3 \end{aligned} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

$$\Rightarrow R_1 \leftarrow R_1 + R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (8) \Rightarrow \text{RREF}$$

